



Identifying price bubble periods in the Bitcoin market-based on GSADF model

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Abstract

The advent of cryptocurrency era is signified by the launch of Bitcoin, which has been considered as a milestone event. Ever since its emergence, Bitcoin has experienced rapid price increase along with great price fluctuations. This paper aims to explore the beginning and the culmination of possible speculative bubbles in the Bitcoin market. Based on the daily closing price of Bitstamp, a major Bitcoin exchange platform, from September 13, 2011 to October 12, 2020, this paper adopts GSADF (generalized sup augmented Dickey–Fuller) to explore the existence of the bubble in the Bitcoin market, estimate the time of emergence and collapse of the bubble, investigate the causes of the bubbles based on its time order, and attempts to put forward some policy suggestions. Through this study, we have found that: (1) There are a lot of speculative bubbles. During the research period, there were a total of four long bubbles (i.e. 2013.01.25–2013.04.05; 2013.11.05–2013.12.10; 2017.04.27–2017.07.05; 2017.07.15–2018.01.09) and four short bubbles (2016.06.08–2016.06.18; 2016.12.30–2017.01.15; 2017.02.22–2017.03.07; 2019.06.19–2019.06.25). (2) The emergence and burst of bubbles were mostly related to the occurrence of relevant events and policies. (3) Bitcoin market is far from mature. A key policy risk triggering the speculative behavior is policy announcement. The prominent innovation of this paper is that the latest developed bubble detection method GSADF has been employed in the Bitcoin market. This method can detect the existence of multiple Bitcoin bubbles and their durations.

Keywords Bitcoin · Multiple bubbles · Generalized sup augmented dickey–fuller (ADF) test

1 Introduction

Bitcoin became the world's fastest growing and most widely used virtual currency. Bitcoin initially got popularity among a small group of geeks as an innovative attempt in cryptography. Nowadays, Bitcoin has gradually entered the public view, and an increasing number of merchants turn to approve it. The emergence and development of Bitcoin has attracted the

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Fig. 1 Bitcoin Price Trend Chart from Sep. 13, 2011 to Dec. 31, 2016 (Units: BTC/USD)

attention of governments and central banks around the world. Bitcoin was originated from a paper of Nakamoto (2008), in the name of *Bitcoin: A Peer-to-Peer Electronic Cash System*. In this paper, Nakamoto described a completely point-to-point electronic cash system, which enabled all payments to be made directly by both parties. The traditional payment pattern through a third-party intermediary such as a commercial bank could be totally removed, thus creating a groundbreaking monetary system. Bitcoin is produced by default computing procedure. As the number of mined Bitcoin increases, the pace of mining new Bitcoin slows down. The amount of the Bitcoin that can be mined will reduce by half every four years. At this pace, the number of Bitcoin will reach a ceiling of 21 million by 2140, which avoids the problem of inflation caused by issuing Bitcoin recklessly (Jia 2013). At present, Bitcoin can be used as a currency for barter trades and on online trading platforms. There are two formal approaches to obtain Bitcoin: the first is through a special Bitcoin computing tool called “digger” for “mining”. The so-called “digger” refers to the computers used to mine Bitcoin, which have professional mining chips. The second way of minning Bitcoin is through exchange with national legal tender. Nowadays, a lot of online trading platforms have emerged, receiving the exchange of Bitcoin and legal tender, among which Mt. Gox in Japan is the first and most powerful one, and Bitstamp and BTC-e are the other two major ones.

With the establishment of a series of trading markets since 2011, the price of Bitcoin has begun to rise rapidly. In order to better reflect the price fluctuations of Bitcoin, we divided the time series on Bitstamp trading platform into two segments. The price trend chart of Bitcoin from September 13, 2011 to December 31, 2016 is shown in Fig. 1. The price trend chart from January 1, 2017 to January 1, 2018 is shown in Fig. 2.

A comparative analysis of Figs. 1 and 2 shows that Bitcoin is traded 24 h a day with no limit of fluctuation range, so that its price can fluctuate by hundreds of dollars in one day. The Bitcoin market has been active since 2013. With no price restrictions and 24/7 trading, the rapid rise of Bitcoin’s market price is accompanied by significant price volatility. Throughout 2013, the price of Bitcoin rose more than 90 times, while trading prices rose explosively and then fell precipitously. In 2014, the price of Bitcoin fell sharply as a result of negative news about Bitcoin. In 2015 and 2016, the price of Bitcoin has experienced phased booms and plunges, and rapid fluctuations. The price of Bitcoin has soared since 2017, closing at nearly \$ 20,000 and falling nearly 100% in the following month. By the end of January 2018, Bitcoin’s market price remained nearly \$ 10,000, which was at

a rather high level historically. In the face of its repeated price fluctuations and explosive price increases, the first question to consider is whether the price reflects the actual value of Bitcoin or unveils the price bubble.

It is important to verify whether there's a price bubble in Bitcoin. We may resort to the method of asset-pricing, which regards a bubble as the part of a market price which is higher or lower than the asset's fundamental value (West 1987; Diba and Grossman 1988; Norden 1996). Therefore, it is necessary to determine the fundamental value of an asset first. After considering all the useful information, the present value of payment of an asset can be considered its fundamental value (Taipalus 2012). But it's difficult to determine a value on Bitcoin because it doesn't have recognizable cash flows. Its market fundamentals are not clear yet. Whether it is a currency, or a commodity, or both has yet reached a consensus (Lo and Wang 2014). Without any intrinsic value, Bitcoin's value is largely based on its speculative value. Some argued that Bitcoins are just claims on numbers (Cheung et al. 2015). Deng also believes that the violent volatility of Bitcoin prices makes it difficult to estimate its intrinsic value (Deng 2017). The uncertainty of its intrinsic value makes the test of price bubbles difficult, which challenges the use of scientific methods to screen the Internet asset bubbles in a timely manner. Dowd discovers that there is a large amount of speculative bubbles in the price of Bitcoin (Dowd 2014). But Gunji argues that Bitcoin prices are rational bubbles by testing the Bitcoin prices of the Chinese Yuan, US Dollar, Euro, and Japanese Yen (Gunji 2016). The Bitcoin price theoretically has no fundamental value. As argued by Gurkaynak, the test of price bubbles with different approaches often leads to inconsistent conclusions, and so does Bitcoin (Gürkaynak 2008).

This paper aims to identify Bitcoin multiple bubbles periods (i.e. when the bubbles occur and burst) by a new date-stamping strategy on the basis of the GSADF (generalized sup augmented Dickey–Fuller) test. Based on the estimation results, the paper explains the causes of each bubble. This paper makes two main contributions: (1) The latest developed bubble detection method has been used (Phillips 2012). The advantages of this method are as follows: Firstly, this method can detect the presence of multiple Bitcoin bubbles and their continuous cycle time, and has a high inspection potential. Second, it can fully demonstrate that Bitcoin bubble can be used throughout the study period of evolution. After a critical review on the pros and cons of different methods, the GSADF method provides

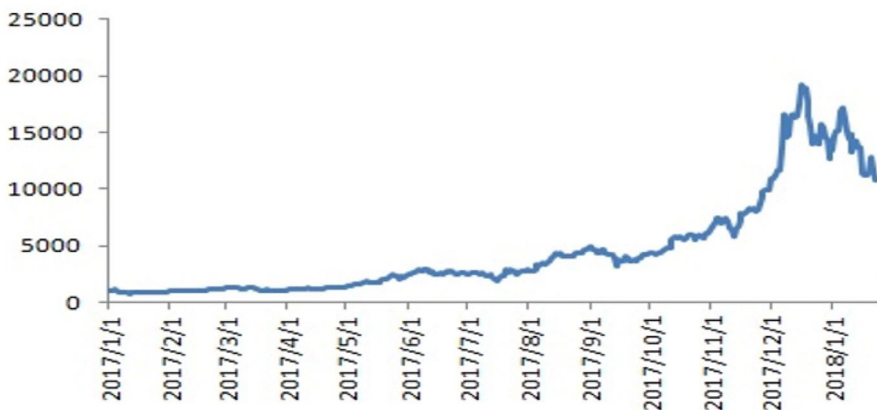


Fig. 2 Bitcoin Price Chart from January 1, 2017 to January 1, 2018 (Units: BTC/USD). Data Source: <https://Bitcoincharts.com/charts/bitstampUSD#tgSzmlg10zm2g25zv>

more reliable results than other alternatives. (2) We identify the beginning and ending time of each Bitcoin market bubble, and conduct in-depth and comprehensive causal analysis.

The rest of this study is organized as follows. Section 2 presents literature review. Section 3 describes the methods. The data source and empirical results are presented in Sect. 4. Finally, conclusions are drawn in Sect. 5.

2 Literature review

Research on Bitcoin has yet emerged until recently (Grinberg 2011; Maurer et al. 2013). Existing literature mostly focuses on the emergence of Bitcoin, its development tendency and supervision strategies (Jacobs 2011; Plassaras 2013). However, there is still no consensus on whether there is a price bubble in the Bitcoin market. More importantly, the present literature on the price bubble of Bitcoin are mainly based on qualitative analysis and related study based on a quantifiable term is needed. The causes of the Bitcoin price bubble and the implications for Internet financial regulation need to be further explored. Woo et al. (2013) discussed the intrinsic value of Bitcoin and they pointed out that market speculation may cause the exchange rate of Bitcoin to fluctuate wildly, thus affecting its degree of social acceptance. Yermack (2013) find that Bitcoin is not so much a currency as a speculative instrument. On the one hand, the daily exchange rate of Bitcoin has no correlation with other major currencies, making it impossible for Bitcoin to be used for risk management purposes. It is also hard for Bitcoin holders to hedge their positions. On the other hand, Bitcoin cannot be used to account for consumer credit or other loan contracts, nor can it be incorporated into a banking system with a deposit insurance feature. Šurda (2012) believes the trust between economic actors ensures the stability of the Bitcoin market. But Bitcoin is neither intrinsically valuable nor representative. Its value depends on market forces and consumer confidence. Therefore, the value of Bitcoin is not guaranteed, and can't get any monetary authority's support. At the same time, it also means that no other factors could control its value and supply except consumer confidence. Through analyzing the features of Bitcoin itself, Ma thinks that the flaws in Bitcoin's design lead to the price speculation (Ma 2014). Combined with the governments' overall negative attitude towards Bitcoin, it's difficult to support its inflated market price. Liao (2014) divides the market development stage of Bitcoin. He argues that Bitcoin transactions have entered a market bubble highlight period which is dominated by speculative demands, and that its market price has severely deviated from its intrinsic value. Based on the theoretical model with positive feedback effect, Hüsler et al. (2013) finds that the positive feedback effect can lead to a price bubble caused by the super-exponential growth of asset prices, which may cause the rapid rise in Bitcoin price (Hüsler et al. 2013). Baek and Elbeck (2015) estimated the volatility of Bitcoin returns and suggest that the Bitcoin market is highly speculative.

3 Theoretical model and methodology

A growing number of studies such as Lucas (1978), Shiller et al. (1984), Tirole (1985), Long and Shleifer et al. (1990) and Gürkaynak (2008) have explored the model of asset pricing. There is a general consensus that bubbles arise when the prices of commodity don't conform to their fundamental values. The model proposed by Gürkaynak is selected as the best method to explore Bitcoin bubble by taking consideration of pros and cons of

different methods. A few assumptions are put forward in this model (Gürkaynak 2008). First, informational asymmetries don't exist. Price fluctuations are not influenced or manipulated by unified traders such as trade unions who capitalize on the information of prices. Second, the typical consumer is risk neutral, which necessarily leads to the assumption that no risk premiums exist. This excludes the risk premiums caused by changes in price or risk, which explains the fluctuation in asset prices. Third, the discount rate is a constant. As mentioned above, the standard model, a special form of a more generic model, permits the emergence of bubbles as shown below.

$$P_t = (1 + r_f)^{-1} E_t(\delta_{t+1} + U_{t+1}) \quad (1)$$

In the equation P_t stands for nature gas price and E_t stands for the expectation in the period t , while δ_{t+1} and U_{t+1} respectively represent the return and invisible component in the period $t + 1$. Then, the equation can be expressed as

$$P_t^f = \sum_{i=0}^{\infty} \left(\frac{1}{1 + r_f} \right)^i E_t(\delta_{t+i} + U_{t+i}), \quad i = 0, 1, 2, \dots, n \quad (2)$$

In this equation, P_t^f represents the fundamental price of nature gas and δ_{t+i} represents the return of nature gas in the period $t + i$. It describes the factors that determine the fundamental price when bubbles don't exist.

$$B_t = (1 + r_f)^{-1} E_t(B_{t+1}) \quad (3)$$

which is any sequence of random variables that meets the homogeneous expectation equation. Therefore, the asset-pricing model can be expressed as

$$P_t = P_t^f + B_t \quad (4)$$

The basic model consists of two parts, a "market fundamental" part, the discounted value of expected future capital gains, and a "bubble" part. In this case, the rational bubble is an essential component of asset prices rather than a mispricing effect. If agents believe that they can sell the asset at a higher price in the future, the asset price is likely to be higher than its fundamental value. It should be noted that the high capital gains expected from asset sales in the future are in accord with no-arbitrage pricing, which are both the value of the right to sell the asset. It's important that there isn't only one path of the bubble (and therefore, the asset price). Equation (4) doesn't limit the law of motion of asset prices except the non-fundamental part. But Eq. (4) provides a different path for every possible value of the bubble's initial level. In order to determine the asset price, an extra assumption about B_t is needed. When $B_t = 0$ is identified, the solution is a special case, which means that the value of the bubble is zero at any time. This fundamental solution forms the basis of the current value-pricing approaches to asset prices, which is also known as "the standard model". Except the absence of bubbles, the assumptions embodied in the formulation of the current value pricing model are effectively expressed.

It is proved by Evans that the traditional tests can't effectively test out bubbles when there is periodic collapse behavior in a time series (Evans 1991). In order to make up for the defect, Phillips et al. use the SADF test, which relies on forwarding recursive regressions connected with sequential right-sided unit root tests, to take explosive behavior into account (Phillips et al. 2011). Through the sequential tests, the influence of unit root behavior on the explosive alternative is proved. It turns out that the SADF is effective in

detecting periodic crashes (Homm and Breitung 2012). In the case of a possible bubble burst, this test is effective for structural breaks. In addition, the SADF test is used not only to investigate financial assets, but also to study physical assets. Meanwhile, Homm and Breitung (2012) also prove that the test of Phillips et al. (2011) is better than others in the study of bubbles. Phillips et al. suggest a forward recursive test procedure and carry out right-side ADF and sup tests (Phillips et al. 2011). They follow the equation as below:

$$Y_t = dT^{-\eta} + \theta y_{t-1} + \varepsilon_t \quad (5)$$

In this equation, d is a constant, T is the sample size, $\eta > 1/2$, $\varepsilon_t \sim \text{NID}(0, \theta^2)$, and $\theta = 1$. A random walk process with an asymptotically negligible drift is taken into account in Eq. (5). We assume that r_1 is the starting point, r_2 is the ending point, in the regression sample, r_w is the window size and $r_2 = r_1 + r_w$. The regression equation forms as follows:

$$\Delta y_t = \alpha_{r_1, r_2} + \beta_{r_1 + r_2} y_{t-1} + \sum_{i=1}^k \varphi_{r_1, r_2}^i \Delta y_{t-i} + \varepsilon_t \quad (6)$$

In this equation, k presents the lag order and $\varepsilon_t \sim \text{NID}(0, \varphi_{r_1, r_2}^i)$. $T_w = [T_{r_w}]$ shows the number of samples in the regression and $[\bullet]$ presents the integer part. $ADF_{r_1}^{r_2}$ shows the ADF statistics in terms of above regression. It is argued by Campbell and Perron (1991) that the lag order k is dictated by significance tests. $H_0 : \beta = 1$ is the null hypothesis of the unit root. At the same time, $H_1 : \beta > 1$ is assumed for the right-tailed unit root.

The SADF test repeatedly assesses the ADF model in terms of a forward-expanding sample sequence. And it also examines the assumption by the sup value of the relevant statistical time series. The window size r_w ranges from r_0 to 1. r_0 represents the smallest window size, while 1 represents the largest window size, which is equal to the full sample. Usually, the starting point r_1 is settled at 0. The equation $r_2 = r_1 + r_w$ shows that the ending point r_2 is contingent on r_w . We also argue $r_2 = r_w$. Therefore, $ADF_0^{r_2}$ statistic shows the the range of sample from 0 to r_2 . The SADF statistic is defined as $\sup_{r_2} \in [r_0, 1] ADF_0^{r_2}$ and denoted by $SADF_{r_0}$.

The SADF method is effective for identifying bubbles when there is a single bubble in the sample. However, several bubbles may exist in the time series if the sample time is long enough. At this point, the SADF test is not appropriate for studying bubbles. Phillips et al. (2013) prove that this method would lose the power to explain the existence of bubbles when there are multiple periods of prosperity and disruption in the time series. This defect becomes apparent when the sample is long enough to have two or more bubbles. To overcome this drawback, the GSADF test is adopted contingent upon flexible window size (Phillips et al. 2013). In comparison to the SADF test, the GSADF method employs a flexible window size through altering the starting and ending points of the recursion instead of settling the starting point, which enlarges the coverage of the sample. Because its window size is more flexible, the GSADF test embodies more subsamples, which enables it to be more effective in identifying the explosive behavior than the SADF method when there are multiple periods in the time series. Taking advantages of supporting multiple bubbles detection and long sample period, GSADF test has been widely used in the research of energy economics and housing markets. Wei et al. (2020) applied GSADF to detect the regional housing bubbles in 140 US cities from 1989 to 2017. Their findings showed that 45 cities had housing bubbles, which frequency ranged from 1 to 8. Li et al. (2020) identified the beginning and the culmination of speculative bubbles in three key regional markets, namely, the European Union (EU), Asia, and the United States (US). The sample

period from January 1996 to June 2017. They found six bubbles appeared in Asia, five in the US, and two occurred in the EU market.

The GSADF method repeatedly runs ADF test regression as Eq. (6) in a sample sequence. However, the sequence of the sample is more extensive than the ADF test. In the GSADF test, the starting point varies from 0 to $r_2 - r_0$, and the ending point r_2 ranges from r_0 to 1. The GSADF test embodies more subsamples; as a result, it has better performance than the SADF test in terms of accounting for the explosive behavior in a few periods. By comparing this test with SADF tests on the basis of their size and power, the remarkable performance of the GSADF test is demonstrated. It is argued by Phillips et al. (2013) that the statistical data of GSADF test are the maximum of ADF statistics with possible starting point and ending point proposed by r_1 and r_2 (Phillips et al. 2013). At the same time, the statistic is indicated by $GSADF(r_0)$ as follows.

$$GSADF(r_0) = \sup_{r_2 \in [r_0, 1], r_1 \in [0, r_2 - r_0]} \{ADF_{r_1}^{r_2}\} \quad (7)$$

An intercept is included in the regression model. The null hypothesis is a random walk with an asymptotically negligible drift (i.e., $dT^{-\eta}$ with $\eta > 1/2$ and constant d). The limit distribution of the GSADF test statistic can be expressed as follows:

$$ADF_{r_1}^{r_2} = \left\{ \frac{\left(\frac{1}{2} \right) r_w [w(r_2)^2 - w(r_1)^2 - r_w] - \int_{r_1}^{r_2} w(r) dr [w(r_2) - w(r_1)]}{r_w^{1/2} \left\{ r_w \int_{r_1}^{r_2} w(r)^2 dr - \left[\theta \int_{r_1}^{r_2} w(r) dr \right]^2 \right\}^{1/2}} \right\} \quad (8)$$

In this equation, $r_w = r_2 - r_1$. It follows a standard Wiener process, the increments of which are independent of those of the distribution $P(s_2) - P(s_1) \sim N(0, s_w)$. If the process is a random walk, the statistics of SADF and GSADF follow standard normal distribution. Statistics of both tests present an asymptotically effective test. The asymptotic critical values of the ADF statistic is derived by the Monte Carlo simulation, which demonstrates that the process follows a random walk. As the process is continuous and random, it would only produce a limited number of finite points. We assume that the intervals such as $n_1, n_2 \dots n_N$ are equidistant. In each interval, a Gaussian random variable would be produced, with a mean of 0 and variance. The right-tail critical values of the GSADF test are larger than those of the SADF test. At the same time, the asymptotic critical values are obtained through a number of simulations, in which the method of bootstrap is applied to the limited sample distributions of the SADF and GSADF tests. The Monte-Carlo simulation method confirms the effectiveness of the bootstrap method and proves that it can identify the explosive processes. Particularly, Pavlidis et al. suggest that the process doesn't have to in accord with the basic principles in this method (Pavlidis et al. 2012).

4 Data and empirical results

This section consists of two parts. The Sect. 4.1 is the description of the data source and descriptive statistics, and the Sect. 4.2 provides the empirical results of eight bubble detections and the analysis of the causes.

Table 1 Descriptive statistics of Bitcoin prices (Units: BTC/USD)

	<i>n</i>	Mean	Max	Min	SD	Skewness	Kurtosis
Bitcoin	3318	3044.62	19,187.78	2.24	3937.66	1.14	3.13

Table 2 SADF and GSADF test results of Bitcoin prices

Test Method	Statistical Value	Critical Level	90%	95%	99%
SADF	23.47***	Critical value of SADF	1.19	1.36	1.54
GSADF	23.47***	Critical value of GSADF	1.97	2.16	2.75

Indicates the significance level $p < 0.01$; * indicates the significance level $p < 0.001$

4.1 Data source

This paper uses the daily closing price of Bitstamp, one of the largest Bitcoin exchange platforms in the world, as the transaction price of Bitcoin. The unit is dollar. The data spans the period between November 13th 2011 and October 12nd 2020. There are 3318 daily observations in the sample period. There are two main reasons for using these data as the Bitcoin prices: (1) Bitstamp is one of the largest Bitcoin trading platforms in the world, with complete data and good representativeness. (2) The US dollar market is an important market for international Bitcoin exchanges. The descriptive statistical results of the data are shown in Table 1.

As can be seen from Table 1, the gap between the highest and lowest prices in the Bitcoin market is obvious. The maximum is 19187.78. The minimum is 2.24. And the maximum is 8566 times the minimum, which shows that the price of Bitcoin in the dollar market fluctuates wildly. The mean value is 3044.62 and the standard deviation is 3937.66, which shows that Bitcoin price variables are too fragmented. The skewness is 1.14 and the kurtosis is 3.13, which seriously deviate from normal distribution. The price of Bitcoin fluctuates wildly and there may exists bubbles. Next section we will conduct a bubble test of the price of Bitcoin.

4.2 Bubble test

Table 2 shows the SADF and GSADF statistical values and critical values of Bitcoin prices. The results are obtained by the 2000 Monte Carlo simulations. The statistical values of SADF and GSADF of Bitcoin prices are both 23.47, which are both larger than the critical value of 99% ($23.47 > 1.54$; $23.47 > 2.75$). It proves that there was a bubble in the dollar market between September 2011 and October 2020.

We tested out the Bitcoin price bubble in the dollar market through SADF and GSADF. Since the GSADF test is more sensitive than the SADF test in the discovery of continuous bubble (Phillips 2012) and the sample data has passed the GSADF test, we only compared GSADF's statistical value sequence with the 99% critical value sequence in order to determine the beginning and burst time of bubbles. The test results are shown in Fig. 3.

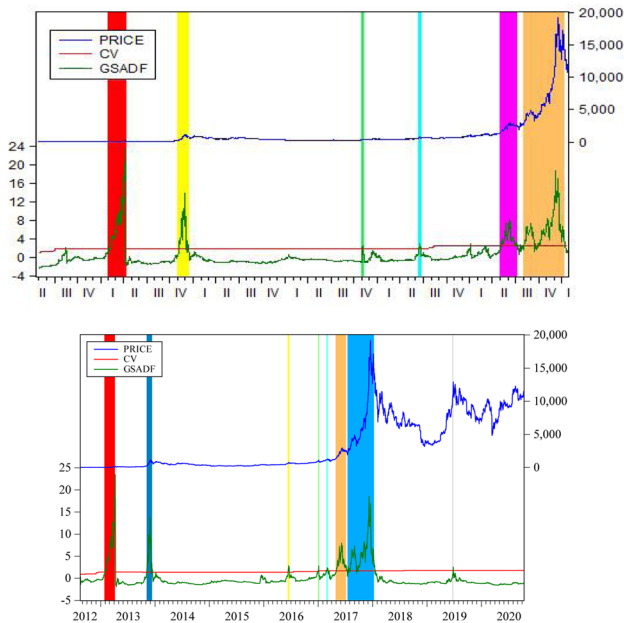


Fig. 3 The Bubble Intervals of Bitcoin Price (Nov. 03, 2011-Oct.24, 2020; Units: BTC/USD). Note: different colors represent different bubble ranges

As shown in Fig. 3, we found that the price of Bitcoin in the international market during the inspection period had a total of eight bubble intervals. The first interval was from January 2013 to April 2013, and the second interval was from November 2013 to December 2013. The third bubble, which began in early June 2016, burst in just ten days. The fourth bubble, which appeared in December 2016 and ended in January 2017. The fifth bubble began in February 2017 and burst in early March 2017. The sixth bubble interval continued from April 27th, 2017 to July 5th. In ten days, the seventh bubble appeared and lasted till January 2018. The last interval was from June 19th, 2019 to June 25th. By observing the duration and form of different bubbles, this paper further analyzes the causes of the generation and bursting of these bubbles.

(1) The first stage (2013.01.25–2013.04.05)

The first significant price increase of Bitcoin began in December 2012 with the birth of the French Central Exchange of Bitcoin. It is the world's first officially sanctioned Bitcoin exchange, which makes Bitcoin no longer limited to transactions within a small range and gives it the necessary condition of becoming a financial product. In March 2013, FinCEN, the United States Department of Treasury Financial Crimes Enforcement Network, issued the regulation on personal management of virtual currency, which recognized Bitcoin as a virtual currency and encouraged the rise of Bitcoin price. With the help of the United States and the French government, the price of Bitcoin jumped by 2300% in five months. With the speculative fever fading, the market

returned to reason, the price of Bitcoin went down. And the speculative bubble soon burst.

(2) The second stage (2013.11.05–2013.12.10)

In October 2013, the drug trafficking website Silk Road was raided by the FBI. The site performed trade by means of virtual currency Bitcoin, which was pegged to the dollar. Bitcoin was the only recognisable currency for people who buy drugs online. The closure of Silk Road caused a wave of panic selling in the Bitcoin industry. The value of Bitcoin rapidly declined and then rebounded quickly, which suggested that there was speculation when Bitcoin price was going down. The issue of Bitcoin regulation was discussed at a US Senate hearing on November 18. Representatives of United States Department of Justice and U.S. Securities and Exchange Commission (SEC) announced Bitcoin was not an illegal currency but a legitimate financial instrument that would benefit the financial system, which aroused strong reaction from the market. On the same day, Bitcoin China announced that they obtained a \$5 million A-round financing. These good news led to a flood of money pouring into the market. The price of Bitcoin continued to shoot up, and even exceeded the price of gold, which ran counter to the demand of keeping a stable value as a currency. On December 3, 2013, People's Bank of China and MIIT (Ministry of Industry and Information Technology of the People's Republic of China) and other three departments jointly issued a notice on the prevention of Bitcoin risks. It defined the identity of Bitcoin and pointed out that Bitcoin was not issued by the monetary authorities, had no legal status equal to the currency, and cannot and should not be used as currency in the market. South Korea's financial regulator announced on December 10th that Bitcoin was not in line with the regulatory rules governing currency transactions by Internet or commercial institutions, and that South Korea would not admit Bitcoin's legal status as currency. With the release of the bad news, the price of Bitcoin plunged. At this stage, the impact of good news was much stronger than bad news, which suggested that there were probably bubbles developed through the boom and plunge of Bitcoin price.

(3) The third stage (2016.06.08–2016.06.18)

There were mainly two reasons contributing to the third boom of Bitcoin. On the one hand, the prosperity of Blockchain technology was the driving force for this price surge. Thanks to the maturity of the blockchain technology behind Bitcoin, a large number of giant banks and financial institutions such as Morgan Stanley, PayPal and MasterCard began to focus on and invest in Bitcoin. The companies saw that blockchain technology could not only be used for Bitcoin, but also for other areas of finance. The features of blockchain can improve the business process and management such as securities, derivatives and loans by this technology. Such expectations had a certain degree of support for Bitcoin demand expectations. Countries also legislated to regulate Bitcoin. The Commodity Futures Trading Commission (CFTC) identified Bitcoin as a bulk commodity and issued a license for Coinbase, a Bitcoin exchange. European Luxembourg also issued a Bitcoin exchange license for trading platform Bitstamp in March 2016. Meanwhile, Japan also proposed to issue the operation standards of Bitcoin. The release of policies popularized Bitcoin to a larger group, creating a good social environment for Bitcoin's development and making Bitcoin popular in the market once again. On the other hand, the 415,000th blocks of Bitcoin had been mined at this time. There were less than 5000 blocks waiting to be mined before the second production halving. According to the Bitcoin open source software agreement, the block rewards for Bitcoin will be halved for every 210,000 blocks. The phenomenon of "rare is pre-

cious” caused by the halving of production broke the previous balance of supply and demand, and triggered a market panic in the supply crunch.

- (4) The fourth stage (2016.12.30–2017.01.15) and the fifth stage (2017.02.22–2017.03.07)

The fourth bubble and the fifth bubble have similarities in many aspects. Therefore, these two stages will be analyzed together.

The occurrence of the fourth bubble is closely related to the rise of Bitcoin in the Chinese market. In 2014–2017, the proportion of Chinese exchanges trading volume in global exchanges increased by 100 times (Gerlach et al. 2019), and the Chinese market increased demand for Bitcoin. At the same time, as the Renminbi depreciated, Chinese investors turn to the blockchain market in pursuit of asset appreciation. In January 2017, the People’s Bank of China foraged about one of the major trading platforms “Bitcoin China” and conducted on-site inspections. These actions signs government policy stances to prevent financial risks caused by abnormal Bitcoin price fluctuations and maintain financial stability. The fifth bubble burst when the SEC rejected the Bitcoin Exchange Traded Fund (ETF) application.

ETF is an open-end fund listed on an exchange with variable fund shares. Bitcoin ETF allows investors to hold digital currencies in a simple way, without the complicated process of buying currencies. Purchasing Bitcoin ETF is per se to indirectly investing in Bitcoin. Since ETF holders do not hold Bitcoin itself, the storage and security issues of digital assets will not be introduced. However, the Bitcoin market is still immature. After three years of review, the SEC finally rejected the Bitcoin ETF submitted by the Winklevoss brothers. The two bubbles were short in duration, and both burst under the restriction of official authorities.

- (5) The sixth stage (2017.04.27–2017.07.05)

The price fluctuations of Bitcoin are determined by complex market factors, including the confidence of the investors, the supply of market level, the influence of the policies and the recognition of its core value. This Bitcoin market bubble was mainly influenced by global good policies. Since entering the second quarter, Japan, Russia, the United States and other places released some policies which were favorable to Bitcoin. In April 27th, the SEC decided to review the Bitcoin ETF. The Bitcoin international market was beginning to rekindle market expectations that Bitcoin ETF would be approved by U.S. regulators. The tax-free trade policy in Japan also allowed the payment properties of Bitcoin to be recognized broadly, reinforcing the market’s confidence in Bitcoin. The Florida government intended to define Bitcoin as a financial instrument. The Russian Federation of Information Technology and Communications announced plans to legalize the blockchain by 2019. In addition, Australia would treat Bitcoin as a currency on July 1st and would abolish the goods and services tax (GST) of Bitcoin, which meant that Australian Bitcoin traders and investors would no longer be taxed on buying and selling bitcoins through regulated exchanges and trading platforms. This news was directly favorable for Bitcoin. Good news led to great demands of investment. On the one hand, some Bitcoin holders reduced payments and hoarded goods, which reduced the demand for Bitcoin transactions and increased the demand for investment. The rise of Bitcoin price became a reality. On the other hand, investors who were sitting on the sidelines started to enter the market, which increased the demand for investment and further pushed up the price of Bitcoin. However, in early July, affected by the forking risk of Bitcoin, the Bitcoin price was sharply lowered, and eventually the bubble burst.

- (6) The seventh stage (2017.07.15–2018.01.09)

The longest price fluctuations of Bitcoin are driven by three reasons. First of all, at the end of July, the price of Bitcoin was steadily recovering after the successful activation of the Bitcoin expansion plan Bitcoin Improvement Proposal 91 (BIP91). At the same time, the Bitcoin Segregated Witness expansion program had been locked and was about to be activated. On August 8, in 479707th block, segregated witness was formally locked and was activated soon after. The number of transactions that could be accommodated in a single block of Bitcoins increased, and the problem of Bitcoin network transfer congestion was alleviated. Meanwhile, the segregated witness could repair transaction malleability, allowing the Bitcoin blockchain to add more new technologies, such as the lightning network. Behind the surge was the important technical power which brought investors a wonderful prospect and strengthened their investment confidence in bitcoin. The price of Bitcoin hit new highs naturally. Secondly, with the introduction of Bitcoin policies in various countries, more and more stores accepted the payment of Bitcoin, and the recognition of Bitcoin was getting higher and higher among investors. The demand for transactions increased, but the number of bitcoins in the market was fixed, which led to higher prices. Thirdly, everyone can become an issuer, a trader, without regulation. Due to its limited amount, Bitcoin has a strong speculative property. With the influx of bitcoin players, the price of Bitcoin has soared far beyond the value of Bitcoin, creating speculative bubbles.

(7) The eighth stage (2019.06.19–2019.06.25)

On June 18, 2019, Facebook released the white paper of “Libra”. Libra is determined to create “a simple global payment system and financial infrastructure” (Libra-Association). In the underlying architecture, Libra adopts a permissioned chain. In terms of economic mechanism, Libra introduces the concept of “The Libra Reserve”, using high-quality liquid assets such as deposits and short-term government bonds constitute Libra Reserves. This reserve aims to provide full guarantees and long-term currency stability. Libra has created a new currency payment business model and subverted the state and bank-core system (Guo and Wang 2020). The announcement of Libra has increased market confidence of Bitcoin, stroke the eighth Bitcoin Bubble. However, the response from regulators and politicians are not optimistic. Bruno Le Maire, the French Finance Minister, said that “Libra could not be allowed to become a sovereign currency, and would require strong consumer protections” (Lesaffre 2018). Maxine Waters, Chairperson of the United States House Committee on Financial Services Committee asked Facebook to stop the development and launch of Libra, and declared “the cryptocurrency market currently lacks a clear regulatory framework” (Wong 2019). The U.S. House Committee on Financial Services Democrats made the same protest, for concerns of privacy, national security, trading, and monetary policy (Services 2019). The eighth bubbles burst as the criticizes and suspicions from authorities.

The causes and mechanism of a series of major economic events, such as the speculative bubble of Bitcoin and the market fluctuation, are complicated and difficult to distinguish. The randomness and overlap of the actual events determine the non-uniformity of the causes and process. In the current situation, the price bubbles of Bitcoin are related to economic, policy, legal and communication factors, and the underlying causes are complex and diverse. Bitcoin’s use value is still low and is highly speculative. In general, boom and plunge of Bitcoin is prone to policies and related events, which are closely related to investors’ speculation. Buyers of Bitcoin flock to the market when where is good news, yet retreat when otherwise, fully reflecting the herd behavior. While good news could spur soaring Bitcoin price, Bitcoin price could also tumble down at bad news.

5 Conclusion

With growing interests and popularity (Frisby 2014; Vigna and Casey 2015), Bitcoin, particularly the bubble features, has yet been fully explored. This paper aims to identify Bitcoin multiple bubbles periods. A recently developed technique by Phillips et al. (2013) has been proven to be effective in detecting multiple bubbles. Random explosive behavior of a given time series can be detected by this technique. This study carry out an econometric research of the existence of bubbles in the Bitcoin market. The estimation results reveal several important findings:

- (1) Over the period 2011–2020, four long bubbles and four short bubbles have been detected, which are mainly driven by socio-economic events. The price of Bitcoin fluctuates frequently and violently due to the influence of market expectation triggered by the release of policy and speculative activities. The generation and bursting of these bubbles seems to coincide with certain major events that occurred in the Bitcoin market.
- (2) Bitcoin market is far from mature. The prices are highly volatile. The abnormal peaks and drops in price emerged frequently. The price is sensitive to policy information, which incurs the policy risk and speculative behavior. However, the potential to recover from drawdowns is equally sharp. Bitcoin is defined as a virtual commodity instead of currency, which is conducive to the return of bitcoin price to rationality. Bitcoin is subject to its own limitations and is unlikely to become a world currency. In case of the questioned stability of Bitcoin price, exerting its monetary value function is waiting to be discussed.

A host of policy recommendations are provided.

First, the national authorities should pay more attention to the verification and restrictions of decentralized financial products. Policies and regulations should be promoted and enacted timely. These regulations will help the investors to perceive the property and investment value of new financial assets, and protect them from speculations. In addition, when the market is conspicuously abnormal, the authorities should announce caution in time, helping investors to conduct rational investment, curb the further aggregation of bubbles, and inhibit the occurrence and spread of speculative bubbles.

Second, Bitcoin should be distinguished from traditional financial products in terms of means of exchange and charge unit. The trading mechanism is not perfect, and the trading platform has certain security risks.¹ For novel decentralized financial products, the authorities should supervise the trading platform, promote the improvement of trading mechanism, prevent the market from being manipulated, and strengthen the regulation of "money laundering" through cryptocurrencies. The loss of investors caused by security problems should be avoided. A fairer and safer investment environment for investors should be established and maintained.

¹ There are many trading platforms of Bitcoin, and there are still some hidden dangers in terms of their security. On February 28, 2014, the world's largest trading platform operators Mt. Bigger at that time declared bankruptcy protection because 850,000 bitcoins (\$467 million) were stolen, which caused a great loss to investors.

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